Zainab Khallouf, Ph.D.

Lecturer in Computer Science & Cybersecurity

zainab.khallouf@univ-ubs.fr | zainab.khallouf@gmail.com
ENSIBS, Université Bretagne Sud, Vannes, France

PHOTO
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PROFILE

With over 20 years of experience in higher education (since 1998), I have developed a strong belief in the transformative power of education. I am currently seeking a permanent academic teaching and research position where I can contribute my expertise in computer science, cybersecurity, distributed systems, and artificial intelligence, while continuing to develop an applied research activity aligned with my long-term academic aspirations.

I hold a Ph.D. in Computer Science from Grenoble Institute of Technology (2006), obtained through an industry-funded CIFRE partnership with France Télécom R&D and INRIA. Since 2021, I have been teaching at ENSIBS, Université Bretagne Sud, where I deliver courses covering web application security, blockchain and Web3, artificial intelligence and neuro-symbolic AI for cybersecurity, distributed and parallel computing for big data, distributed systems, and component-based software engineering. I also serve as the pedagogical coordinator for the fourth-year Cybersecurity and Data Science program, contributing to curriculum development and student mentoring.

My research interests lie at the intersection of artificial intelligence and cybersecurity, with a particular focus on the application of AI to emerging security challenges. I am the author of publications covering multicast routing security, intrusion detection, e-commerce systems, and, most recently, neuro-symbolic AI for blockchain fraud detection, including work conducted in collaboration with IRISA and published at MLN 2025.

Throughout my academic career, including during challenging periods of conflict in Syria, I have remained deeply committed to teaching, knowledge sharing, and student development. I now wish to build on this experience within a stable, collaborative academic environment.

AREAS OF EXPERTISE

Cybersecurity	Web application security (OWASP), network and information security, identity and access management (OAuth 2.0, OpenID Connect), intrusion detection, blockchain security and Web3
Artificial intelligence	Neuro-symbolic AI (Logic Tensor Networks), explainable AI, machine and deep learning for security, anomaly detection, behavioral analysis (UEBA)
Distributed systems	gRPC, MPI, Apache Kafka, microservices, Apache Spark, Hadoop, Dask, Ray, CUDA
Software engineering	Component-based design (Spring, OSGi, Jakarta EE), web development (Node.js, React, Angular)
Languages	Arabic, French, English (languages of instruction)

ACADEMIC POSITIONS

France

Sept. 2025 – present
Contractual Lecturer (CDD-LRU)
ENSIBS, Université Bretagne Sud, France

Nov. 2021 – Sept. 2025
Contractual Lecturer–Researcher (CDD-LRU)
ENSIBS, Université Bretagne Sud, France
Supervision of internships, projects, and work-study (apprenticeship) contracts. See also *Academic Responsibilities*.

Syria – University of Homs (formerly Al-Baath University)

Dec. 2015 – Oct. 2021 **Assistant Professor**

Faculty of Computer Engineering, University of Homs

Vice-Dean for Scientific Affairs from Aug. 2017 (see *Academic Responsibilities*).

June 2010 – Dec. 2015 **Lecturer (full-time)**

Faculty of Computer Engineering, University of Homs

June 2008 – June 2010 **Probationary Lecturer**

Faculty of Computer Engineering, University of Homs

Appointed following the Ph.D. from Grenoble Institute of Technology. Also taught a Master's-level course for two years at the University of Damascus.

May 2006 – June 2008 **Faculty member**

Faculty of Computer Engineering, University of Homs

Resumed duties after obtaining the Ph.D. in France.

Sept. 2000 – May 2006 **State scholarship holder – study mission in France**

Sent to France by the University to obtain the qualification required for a faculty appointment

See *Education*: DESS, DEA and Ph.D. (CIFRE).

Oct. 1998 – Aug. 2000 **Full-time Teaching Assistant**

Faculty of Sciences, Department of Mathematics (Computer Networks specialty), then Faculty of Computer Engineering

Appointed in Sept. 1998 (took up duties 25 Oct. 1998); transferred to the Faculty of Computer Engineering in Jan. 1999. Internal mission to follow a 9-month French language course (Oct. 1999 – June 2000), then sent to France on a study scholarship (July 2000).

ACADEMIC RESPONSIBILITIES

2022–2023 – present **Pedagogical Coordinator, 4th year**

Cybersecurity and Data Science *program, ENSIBS – Université Bretagne Sud*

Student mentoring, follow-up of work-placement evaluations and grading, curriculum development.

From Aug. 2017 **Vice-Dean for Scientific Affairs**

Faculty of Computer Engineering, University of Homs, Syria

EDUCATION

Dec. 2002 – Mar. 2006 **Ph.D. in Computer Science – Systems and Software**

Grenoble Institute of Technology (INPG), France

Funded by the French National Association for Research and Technology (ANRT) through a CIFRE industrial research grant, in partnership with France Télécom R&D (Lannion) and INRIA Rhône-Alpes (Grenoble).
Dissertation: *Multicast Routing Infrastructure Security: The Network Operator's Point of View.*

Sept. 2001 – July 2002 **DEA (Postgraduate Degree) – Computer Science, Systems and Communication**

Joseph Fourier University, Grenoble, France

Research thesis: *A Study of High-Speed Forward Error Correction (FEC) Codes for the Reliable Multicast Delivery of Bulk Data.*

Sept. 2000 – July 2001 **DESS – Complementary Competence in Computer Science**

Joseph Fourier University, Grenoble, France

Sept. 1999 – June 2000 **Intensive French Language Program**

Damascus, Syria

Sept. 1997 – Aug. 1999 **Postgraduate Diploma in Computer Science**

Al-Baath University, Homs, Syria

Final project: *Using Unicode and Java to Solve the Problems of the Arabic Language in Microsoft Internet Explorer.*

Sept. 1993 – June 1997 **Bachelor's Degree in Mathematics, Computer Science Track**

Al-Baath University, Homs, Syria

Three years of applied mathematics, one year of computer science. Graduated with Highest Honors – GPA 80.2/100 (Top Student Award).

TEACHING ACTIVITIES

Over the course of my career I have taught computer science, cybersecurity, and computer networks at undergraduate and graduate level, in Arabic, French, and English, depending on the institution and program.

ENSIBS – Université Bretagne Sud (2021 – present)

From November 15, 2021 to June 2026, I completed approximately **1,824 tutorial-equivalent hours (HETD)** of total service, including teaching, supervision, and administrative responsibilities. The list below gathers all courses taught since 2021, with program, semester, period, language of instruction, and main keywords.

Component-Based Software Engineering *2021–2026 | French*

Software Cybersecurity (Cybersécurité du logiciel) / S8. *Keywords:* Component-based software engineering, fine-grained and coarse-grained component models, tight and loose coupling, cohesion, inversion of control, dependency injection. Case studies: Spring (fine-grained), OSGi (coarse-grained), Jakarta EE (formerly Java Platform, Enterprise Edition).

Introduction to Programming *2021–2026 | French*

PEI-A Preparatory Track, Lorient / S2. *Keywords:* Event-driven programming, MIT App Inventor, visual and block-based programming, programming basics.

Data Analysis for Cybersecurity *2021–2022 | French*

Cyberdefense / S8. *Keywords:* Log analysis, Apache Spark, MLlib, machine learning for intrusion detection, Random Forest, distributed machine learning, Spark-based intrusion detection systems.

Web Programming *2021–2022 | French*

Cyberdefense / S6. *Keywords:* HTML5, CSS3, JavaScript, Node.js, Express.js, React.js.

Web Application Security *2022–2026 | French*

Cybersecurity & Data Science (Cybersécurité et Sciences des Données), Software Cybersecurity / S7. *Keywords:* OWASP Top 10, SQL injection, XSS, CSRF, session management, OpenID Connect, OAuth.

Blockchain Technologies & Web3 *2022–2026 | French*

Cybersecurity & Data Science, Software Cybersecurity / S8. *Keywords:* Distributed ledgers, consensus mechanisms, smart contracts, Ethereum, Solidity, DeFi, NFTs, Web3.

Statistical AI for Cybersecurity *2022–2023 | French*

Cybersecurity & Data Science / S8. *Keywords:* Introduction to machine learning for cybersecurity, spam classification, deep learning, anomaly detection with autoencoders, adversarial machine learning, malware-to-image transformation for classification using deep learning.

Secure Distributed Systems Development *2023–2026 | French*

Software Cybersecurity / S8. *Keywords:* Distributed system design, peer-to-peer systems, Remote Procedure Call (RPC), gRPC, Message Passing Interface (MPI), message-oriented communication, Apache Kafka, microservices, distributed transactions.

Introduction to Research & Technology Watch *2023–2026 | English*

Cybersecurity & Data Science, Software Cybersecurity / S7. *Keywords:* Research methodology, literature review, technology watch.

(Neuro-)Symbolic AI for Cybersecurity *2023–2026 | English*

Cybersecurity & Data Science / S9. *Keywords:* Neuro-symbolic AI, Logic Tensor Networks, explainable AI.

Distributed & Massively Parallel Computing (Big Data) *2023–2026 | French*

Cybersecurity & Data Science / S9. *Keywords:* CUDA, distributed computing, Apache Kafka, Apache Spark, Hadoop, MapReduce, Dask, Ray.

Behavioral Analysis *2024–2026 | French*

Cybersecurity & Data Science / S9. *Keywords:* User and Entity Behavior Analytics (UEBA), anomaly detection, shopping cart analysis.

Random Number Generators *2025–2026 | French*

Software Cybersecurity / S6. *Keywords:* PRNGs, statistical testing.

University of Homs, formerly Al-Baath University, Syria (2006 – 2021)

Total teaching volume exceeds **3,000 hours**. All courses were delivered in Arabic (except *English for Informatics*, delivered in English).

Computer Networks Security *2006–2021 | 664 hrs*

5th year, Networks Dept.. *Keywords:* Authentication services (Kerberos, PKI), PGP, S/MIME, IPSec, IKE, SSL/TLS, firewalls (Cisco ACLs, ASA, IPtables), Snort IDS, wireless network security, spam classification (Naive Bayes).

Information Security *2006–2021 | 448 hrs*

5th year, Software Dept.. *Keywords:* Security services, threats, classical ciphers, symmetric/asymmetric algorithms, key management, Kerberos, SSL/TLS, WS-Security, RESTful web service security (JWT, APIMAN, OAuth 2.0, OpenID Connect).

Information Security *2008–2021 | 352 hrs*

4th year, Networks Dept.. *Keywords:* Security services, threats, classical ciphers, symmetric/asymmetric algorithms, key management, message authentication, Kerberos, SSL/TLS.

E-commerce Systems *2008–2019 | 432 hrs*

5th year, Software & Networks. *Keywords:* Multi-tier e-commerce architectures, shopping cart implementation (Angular + ASP.NET Core 2, Angular + Java EE), Ionic, WordPress.

Advanced Information Security *2013–2021 | 512 hrs*

Master 1, Software Dept.. *Keywords:* RESTful web service security, DKIM, OAuth 2.0, OpenID Connect, Keycloak, federation, SAML, machine/deep learning for security.

Advanced Computer Network Management *2012–2018 | 384 hrs*

Master 1, Networks Dept.. *Keywords:* Network measurement and data analysis, Ruby, Gnuplot, SNMP, querying SNMP devices with Python, Round-Robin Database (RRD).

Advanced Systems & Network Security *2018–2021 | 192 hrs*

Master 1, Networks Dept.. *Keywords:* Security data analysis with R/Python, machine learning and deep learning for security.

Quality & Reliability of Computer Systems *2006–2008 | 96 hrs*

5th year, Software & Networks. *Keywords:* MTTF, MTTR, reliability modelling, Weibull distribution, reliability block diagrams, Markov models, fault-tolerant distributed systems, RAID.

English for Informatics 3 *2008–2010 | 96 hrs*

5th year, Software & Networks. *Keywords:* Technical English for computer science (taught in English).

Distributed Systems *2008–2010 | 80 hrs*

5th year, Software & Networks. *Keywords:* Distributed system architecture, inter-process communication, CORBA, distributed transactions, global time and clocks, distributed mutual exclusion algorithms.

Wireless Networks *2008–2013 | 80 hrs*

4th year, Networks Dept.. *Keywords:* Wireless LANs, MANET routing, NS2, Mobile IPv4, wireless TCP, GSM.

Advanced Computer Network Design *2012–2013 | 64 hrs*

Master 1, Networks Dept.. *Keywords:* Traffic shaping, VLANs, campus network design, redundancy and availability.

Operating Systems *2010–2011 | 48 hrs*

4th year, Networks Dept.. *Keywords:* Processes, inter-process communication, threads, process synchronization, Pthreads, producer-consumer problem.

Computer Ethics & Social Responsibility *2006–2008 | 12 hrs*

5th year, Software & Networks. *Keywords:* Code of ethics, ethical theories, case study analysis, free and open-source software licensing.

Computer Networks Security (Master's course) *2008–2010 | ∼32 hrs*

University of Damascus. *Keywords:* P2P systems security, Chord DHT, P2PSIP, key management for secure group communication, threshold cryptography, secret sharing, MANET security, sensor network security.

MASTER'S THESIS SUPERVISION

Supervised or co-supervised 8 master's theses (2–3 year format) at the Faculty of Computer Engineering, University of Homs, Syria (2014–2021), covering cybersecurity, distributed systems, IoT, and machine learning:

- Intrusion detection system performance analysis and improvement (co-supervisor, 2014)
- Quality of service for routing protocols in mobile networks (co-supervisor, 2015)
- Distributed systems performance modelling via Petri nets (2016)
- Securing hybrid routing protocols in MANETs (2016)
- IoT system for vital-sign monitoring (2019)
- Configuration management protocols (2019)
- Behavioral detection in IoT via deep learning (2021)
- Multi-cloud storage security via secret sharing (2021)

RESEARCH ACTIVITIES

Having concentrated for several years primarily on teaching – including building the Cyber Data curriculum at ENSIBS – I have recently developed a complementary research activity directly connected to my teaching, focused on interpretable machine learning approaches applied to the cybersecurity of digital systems.

Neuro-Symbolic AI for Cybersecurity: Fraud Detection on the Ethereum Blockchain

My recent research applies artificial intelligence and blockchain technology to cybersecurity, in collaboration with Prof. Pierre-François Marteau (IRISA). Although not formally affiliated with a research team for administrative reasons specific to the school, this collaboration produced a publication presented at MLN 2025.

Neuro-symbolic AI (NSAI) combines the pattern-recognition strengths of neural networks with the logical reasoning of symbolic systems. This study explored the application of NSAI, via Logic Tensor Networks (LTNs), to detecting fraudulent activity on the Ethereum blockchain, formulating the task as binary classification distinguishing legitimate from fraudulent transactions. We proposed and evaluated an **ensemble** NSAI framework based on a voting mechanism, applied to a public Ethereum fraud-detection dataset. Experimental results showed the ensemble architecture outperforming approximately 80% of individual models, while remaining fully compatible with Ethereum's peer-to-peer architecture. Current work continues to focus on neuro-symbolic AI for anomaly detection, combining classical and cybersecurity-oriented data analysis in close connection with my teaching activities.

PUBLICATIONS

1. Ensemble Neuro-Symbolic AI and Logic Tensor Networks for Detecting Fraud on the Ethereum Blockchain. Z. Khallouf, P.-F. Marteau. *8th International Conference on Machine Learning for Networking (MLN 2025)*, Paris, France, Dec. 2–4, 2025.

2. Classification of Intrusions in Local Area Networks (LANs) using Random Forests and Apache Spark. Z. Khallouf. *Al-Baath University Journal*, 2021. (In Arabic)
3. Securing Representational State Transfer (RESTful) Web Services Using Open ID Connect. Z. Khallouf. *Al-Baath University Journal*, 2018. (In Arabic)
4. Comparison between AngularJS 1 and Angular 2 in the Implementation of the Presentation Layer of Multitiered E-commerce Applications. Z. Khallouf. *Al-Baath University Journal*, 2017. (In Arabic)
5. Implementing Multi-Tiered E-commerce Applications and Assessing the Impact of Different Techniques on Performance. Z. Khallouf. *Al-Baath University Journal*, 2016. (In Arabic)
6. Simulation of Routing Protocols in Mobile Ad hoc Networks (MANETs). Z. Khallouf. *Al-Baath University Journal*, 2011. (In Arabic)
7. Survey and Taxonomy of Approaches to Improve the Efficiency of Intrusion Detection Systems. Z. Khallouf. *Al-Baath University Journal*, 2010. (In Arabic)
8. Studying the Use of Group Communication Algorithms to Build Fault-Tolerant Services through Replication. Z. Khallouf. *Al-Baath University Journal*, 2010. (In Arabic)
9. Trust Management in Peer-to-Peer SIP Using the Security Assertion Markup Language. Z. Khallouf. *International Conference on Information & Communication Technologies: from Theory to Applications (ICTTA'08)*, Damascus, Syria, Apr. 7–11, 2008.
10. A Filtering Approach for an IGMP Flooding Resilient Infrastructure. Z. Khallouf, V. Roca, R. Moignard, S. Loye. *4th Conference on Security and Network Architectures (SAR'05)*, Batz-sur-Mer, France, Jun. 2005.
11. Design and Evaluation of a Low-Density Generator Matrix (LDGM) Large Block FEC Codec. V. Roca, Z. Khallouf, J. Labouré. *5th International Workshop on Networked Group Communication (NGC'03)*, Munich, Germany, Sept. 2003.

SELECTED PRESENTATIONS & CONFERENCES

- Ensemble Neuro-Symbolic AI and Logic Tensor Networks for Detecting Fraud on the Ethereum Blockchain, Community Cyber UBS-Lorient, Dec. 4, 2025 (English)
- Ensemble Neuro-Symbolic AI and Logic Tensor Networks for Detecting Fraud on the Ethereum Blockchain, MLN'2025, Paris, Dec. 2–4, 2025 (English)
- Applying Neuro-Symbolic AI for Detecting Fraud on the Ethereum Blockchain, Instant Cyber & IA 25, Université Bretagne Sud, May 2025 (French)
- Comparing Approaches to Improve the Efficiency of Intrusion Detection Systems, ICCIS 2010, Egypt (English)
- Data Protection Using Replication Based on Group Communication, 6th ICT Security Forum, Damascus, Syria, Oct. 2010 (English)
- Trust Management in Peer-to-Peer SIP Using SAML, ICTTA'08, Damascus, Syria (English)
- About Integrating Ethics in the Software Engineering Curriculum, SCS Workshop, Damascus, Dec. 2007 (Arabic)
- Secure Multicast Routing Infrastructure: The Network Operator's Point of View, SAR'04, La Londe, Côte d'Azur, France, Jun. 2004 (French)
- A Filtering Approach for an IGMP Flooding Resilient Infrastructure, SAR'05, Batz-sur-Mer, France, Jun. 2005 (English)

OTHER INTERESTS

- **Art:** painting and sculpture. Solo exhibition, UBS, France (2025); two collective exhibitions, Séné, France (2026).
- **Sports:** UBS Relay Marathon (2023, 2024), 10 km.

Last updated: November 2026